Original article

Prognostic significance of gross extrathyroidal extension invading only strap muscles in differentiated thyroid carcinoma

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Background: **In the eighth edition of the AJCC staging system for differentiated thyroid carcinoma (DTC), minimal extrathyroidal extension (ETE) is no longer a determinant of T3 category. Instead, gross ETE invading only strap muscles has been designated as a new T3b category. The long-term prognosis of patients with DTC and gross ETE invading only strap muscles was investigated.**

Methods: **This was a retrospective analysis of patients who underwent thyroidectomy between 1996 and 2005. Differences in cancer-specific and recurrence-free survival according to extent of ETE were assessed.**

Results: **A total of 3174 patients with DTC were included. No significant differences were observed in 10-year cancer-specific survival among patients with no ETE (98·6 per cent), microscopic ETE (98·3 per cent) and gross ETE invading only strap muscles (98·9 per cent) ($P = 0·375$). The 10-year recurrence-free survival rate for patients with gross ETE invading only strap muscles (89·2 per cent) was shorter than that for patients with no ETE (93·7 per cent; $P = 0·016$), but similar to that of patients with microscopic ETE (90·3 per cent). In univariable analysis, patients with gross ETE invading only strap muscles had a significantly higher risk of recurrence than those with no ETE (hazard ratio (HR) 1·67, 95 per cent c.i. 1·10 to 2·55; $P = 0·017$). In multivariable analysis, gross ETE invading only strap muscles was not an independent predictor of recurrence (HR 1·09, 0·71 to 1·69; $P = 0·685$).**

Conclusion: **Although gross ETE invading only strap muscles may provide prognostic information about long-term recurrence, it does not affect mortality. The actual impact of gross ETE invading only strap muscles will be important in revising the staging system in the future.**

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Introduction

Prognostic factors for differentiated thyroid carcinoma (DTC) are well described, and include age at diagnosis, tumour size, regional lymph node metastases and distant metastases[1]. The presence of extrathyroidal extension (ETE) is considered important and is included in almost all prognostic scoring systems as a staging variable[2]. This is, however, based mainly on studies with gross ETE invading adjacent structures as identified by intraoperative findings[3].

On histology, ETE can be categorized by extent of invasion, from gross ETE invading major structures to minimal ETE identified only by microscopy. Since the sixth edition of the AJCC TNM staging system[4] for DTC first used minimal ETE as a staging variable, several studies[5–7] have investigated its prognostic significance. Although there is general agreement that gross ETE invading major structures is independently associated with poor prognosis, there is little evidence to support minimal ETE as an independent predictor of persistence/recurrence or survival. Two reviews[8,9] have reported that minimal ETE has no effect on long-term outcomes.

For this reason, in the eighth edition of the AJCC TNM staging system[10], the presence of minimal ETE no longer constitutes a T3 category. Instead, gross ETE invading only strap muscles is designated as a new T3b category. The new staging system also distinguishes anterior ETE from posterior ETE for the first time. However, a detailed

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analysis of the prognostic significance of gross ETE invading only strap muscles has not yet been undertaken. This study investigated the impact of gross ETE invading only strap muscles on persistence/recurrence and survival, with the goal of establishing its prognostic value in DTC.

Methods

Medical records of patients who underwent thyroidectomy for DTC at a tertiary Korean hospital between 1996 and 2005 were reviewed retrospectively. Only patients with histologically confirmed DTC (papillary thyroid carcinoma, follicular thyroid carcinoma or Hürthle cell thyroid carcinoma) were included; those with poorly differentiated thyroid carcinoma and undifferentiated carcinoma were excluded. The study protocol was approved by the Institutional Review Board of Samsung Medical Centre (2016-05-053). Patient consent was waived in some instances by the committee, because of the retrospective chart review study design and use of only deidentified clinicopathological information.

Patients were categorized into four groups based on the degree of ETE: none, microscopic, gross invading only strap muscles (sternohyoid, sternothyroid, thyrohyoid or omohyoid muscles), and gross invading the subcutaneous soft tissues, larynx, trachea, oesophagus, muscles, recurrent laryngeal nerve or prevertebral fascia, or encasing the carotid artery or mediastinal vessels. The degree of ETE was based on pathology and operative reports. No ETE in this study was diagnosed when completely intrathyroidal tumours were identified by pathological examination. Microscopic ETE referred to the presence of ETE detected only by microscopy (not grossly evident), with tumour involvement of perithyroidal adipose tissue, strap muscles, nerves or small vascular structures.

Data collection and outcomes

Clinicopathological information was obtained from a prospective institutional thyroid cancer database, with each patient having follow-up at least once yearly. Demographic and treatment data and laboratory test/imaging results stored in electronic medical record systems were extracted automatically. The surgical notes, pathological reports and imaging results were coded by the database coordinator. Follow-up was maintained by a database coordinator through review of clinical charts. Survival status and cause of death were ascertained by linkage to national death certificate data from the Korean National Statistical Office.

Thyroid cancer-specific survival (CSS) was defined as time from initial surgery to last follow-up or death caused by DTC. Recurrence-free survival (RFS) was defined as time from initial surgery to last follow-up or development of first evidence of structural recurrence. Disease recurrence in this study referred to structural recurrence, confirmed as the existence of physical tumours of recurrent or persistent disease by standard cytological, histological and radiographic criteria.

Table 1 Baseline characteristics of patients with differentiated thyroid carcinoma

	No. of patients* (n = 3174)
Mean(s.d.) age at diagnosis (years)	46·5(12·3)
Sex	
F	2746 (86·5)
M	428 (13·5)
Method of tumour detection	
Non-palpable (detected by ultrasonography)	2303 (72·6)
Palpable or symptomatic	631 (19·9)
Incidentally detected	3 (0·1)
Unspecified or unknown	237 (7·5)
Histology	
Papillary thyroid carcinoma	3073 (96·8)
Follicular thyroid carcinoma	92 (2·9)
Hürthle cell thyroid carcinoma	9 (0·3)
Primary tumour size (cm)	
≤ 1·0	1518 (47·8)
1·1–2·0	1002 (31·6)
2·1–4·0	535 (16·9)
> 4·0	119 (3·7)
Mean(s.d.)	1·5(1·3)
Extrathyroidal extension	
None	1362 (42·9)
Microscopic	1377 (43·4)
Gross invading only strap muscles	261 (8·2)
Gross invading perithyroidal structures other than strap muscles	174 (5·5)
Cervical lymph node metastases	
No	2059 (64·9)
Yes	1115 (35·1)
Distant metastases	
No	3125 (98·5)
Yes	49 (1·5)
Extent of surgery	
Total thyroidectomy	2921 (92·0)
Lobectomy	253 (8·0)
Lymph node dissection	
No	1213 (38·2)
Central compartment neck dissection	1473 (46·4)
Modified radical neck dissection	488 (15·4)
Radioiodine ablation	
No	452 (14·2)
Yes	2722 (85·8)

*With percentages in parentheses unless indicated otherwise.

Statistical analysis

Continuous variables are presented as mean(s.d.), and categorical data as numbers with percentages. ANOVA and

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Table 2 Characteristics of patient groups stratified by degree of extrathyroidal extension

	No ETE (*n* = 1362)	Microscopic ETE (*n* = 1377)	Gross ETE invading only strap muscles (*n* = 261)	Gross ETE invading perithyroidal structures other than strap muscles (*n* = 174)	*P*†	*P*‡
Age at diagnosis (years)*	45·5(11·6)	46·2(12·2)	49·0(13·1)	52·4(15·0)	< 0·001	< 0·001
Sex					0·834	0·653
F	1170 (85·9)	1199 (87·1)	227 (87·0)	150 (86·2)		
M	192 (14·1)	178 (12·9)	34 (13·0)	24 (13·8)		
Histology					< 0·001	< 0·001
Papillary thyroid carcinoma	1291 (94·8)	1354 (98·3)	259 (99·2)	169 (97·1)		
Follicular thyroid carcinoma	63 (4·6)	23 (1·7)	2 (0·8)	4 (2·3)		
Hürthle cell thyroid carcinoma	8 (0·6)	0 (0)	0 (0)	1 (0·6)		
Primary tumour size (cm)*	1·25(1·10)	1·53(1·19)	1·91(1·35)	2·42(1·87)	< 0·001	< 0·001
Cervical lymph node metastases					< 0·001	< 0·001
No	1066 (78·3)	801 (58·2)	128 (49·0)	64 (36·8)		
Yes	296 (21·7)	576 (41·8)	133 (51·0)	110 (63·2)		
Distant metastases					< 0·001	0·397
No	1349 (99·0)	1356 (98·5)	258 (98·9)	162 (93·1)		
Yes	13 (1·0)	21 (1·5)	3 (1·1)	12 (6·9)		
Extent of surgery					< 0·001	< 0·001
Total thyroidectomy	1186 (87·1)	1311 (95·2)	257 (98·5)	167 (96·0)		
Lobectomy	176 (12·9)	66 (4·8)	4 (1·5)	7 (4·0)		
Lymph node dissection					< 0·001	< 0·001
No	654 (48·0)	492 (35·7)	35 (13·4)	32 (18·4)		
Central compartment neck dissection	587 (43·1)	647 (47·0)	169 (64·8)	70 (40·2)		
Modified radical neck dissection	121 (8·9)	238 (17·3)	57 (21·8)	72 (41·4)		
Radioiodine ablation					< 0·001	< 0·001
No	338 (24·8)	78 (5·7)	15 (5·7)	21 (4·6)		
Yes	1024 (75·2)	1299 (94·3)	246 (94·3)	153 (87·9)		

Values in parentheses are percentages unless indicated otherwise; *values are mean(s.d.). ETE, extrathyroidal extension. †Comparison of four groups; ‡comparison of three groups: no ETE, microscopic ETE and gross ETE invading only strap muscles (χ^2 test for categorical data and ANOVA for continuous data).

χ^2 tests were used to assess relationships between continuous and categorical variables respectively. Thyroid CCS and disease recurrence probabilities were estimated by the Kaplan–Meier method, with curves compared between groups using log rank tests. Cox regression and Cox proportional hazards analyses were used to assess clinicopathological factors associated with thyroid cancer-related deaths or structural recurrences. To address the possible impact of size of the nodule and change in patient characteristics and clinical practice, in subgroup analyses the cohort was stratified according to a tumour size cut-off of 4 cm, as used for T2 and T3a criteria in the AJCC TNM system, and the year of initial surgery in 5-year intervals. $P < 0·050$ was considered statistically significant. Statistical analysis was done using SPSS® version 23.0 (IBM, Armonk, New York, USA).

Results

A total of 3174 patients (2746 women and 428 men), with a mean age of 46·5 years, were included. Overall median follow-up from initial surgery to death or recurrence or censoring was 12·4 (i.q.r. 11·1–14·5) years. Clinicopathological characteristics are shown in *Table 1* and *Table S1* (supporting information). Tumours in 2306 patients (72·7 per cent) were non-palpable or detected incidentally. The majority of patients (3073, 96·8 per cent) had papillary thyroid carcinoma and 101 (3·2 per cent) had follicular thyroid carcinoma (including Hürthle cell thyroid carcinoma). The mean size of the primary tumour was 1·5 cm.

Of these 3174 patients, 1362 (42·9 per cent) had no ETE, 1377 (43·4 per cent) had microscopic ETE only, 261 (8·2 per cent) had gross ETE invading only strap muscles, and 174 (5·5 per cent) had gross ETE invading perithyroidal structures other than strap muscles. Comparison of clinicopathological characteristics among these four groups is shown in *Table 2*. Patients with either microscopic ETE or gross ETE invading only strap muscles tended to have larger tumours and more lymph node metastases, and were more likely to undergo total thyroidectomy, lymph node dissection and radioiodine ablation than patients with no ETE (all $P < 0·001$). The rates of distant metastasis did not

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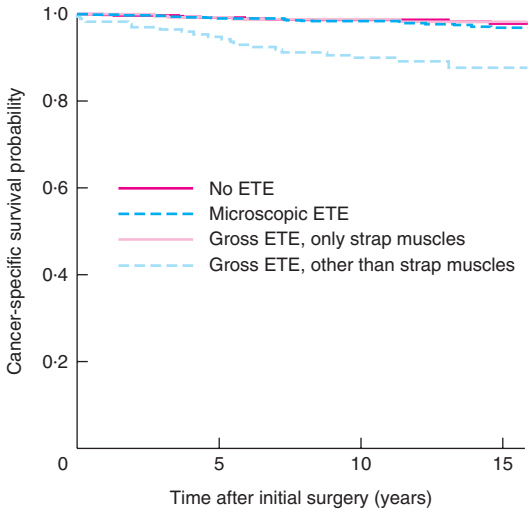
Table 3 Thyroid cancer-related deaths and cancer-specific survival according to degree of extrathyroidal extension

			Cancer-specific survival (%)		Cox regression analysis	
	No. of patients	No. of deaths*	10 years	15 years	Hazard ratio†	*P*
No ETE	1362	25 (1·8)	98·6	98·0	1·00 (reference)	
Microscopic ETE	1377	35 (2·5)	98·3	96·9	1·14 (0·84, 2·33)	0·203
Gross ETE invading only strap muscles	261	6 (2·3)	98·9	98·2	1·49 (0·61, 3·63)	0·385
Gross ETE invading perithyroidal structures other than strap muscles	174	19 (10·9)	90·0	87·6	6·63 (3·65, 12·01)	< 0·001
Overall	3174	85 (2·7)				

Values in parentheses are *percentages and †95 per cent confidence intervals. ETE, extrathyroidal extension.



No. at risk				
No ETE	1362	1335	1228	343
Microscopic ETE	1377	1354	1246	294
Gross ETE, strap muscles only	261	259	235	20
Gross ETE, other than strap muscles	174	161	144	29

Fig. 1 Kaplan–Meier survival curves for thyroid cancer-specific survival according to extent of extrathyroidal extension (ETE). $P = 0·201$, no ETE *versus* microscopic ETE; $P = 0·317$, no ETE *versus* gross ETE invading only strap muscles; $P < 0·001$, no ETE *versus* gross ETE invading perithyroidal structures other than strap muscles; $P = 0·778$, microscopic ETE *versus* gross ETE invading only strap muscles; $P < 0·001$, microscopic ETE *versus* gross ETE invading perithyroidal structures other than strap muscles; $P < 0·001$, gross ETE invading only strap muscles *versus* gross ETE invading perithyroidal structures other than strap muscles (log rank test)

differ between the groups with no ETE, microscopic ETE and gross ETE invading only strap muscles ($P = 0·397$). Patients with gross ETE invading perithyroidal structures other than strap muscles tended to have larger tumours, more lymph node metastases and significantly more distant metastases compared with patients in the other three groups (all $P < 0·001$).

Cancer-specific survival according to extent of extrathyroidal extension

Of the 3174 patients, 85 (2·7 per cent) died from thyroid cancer a median of 8·8 (i.q.r. 3·3–11·2) years after the initial operation (*Table 3*). No significant differences in 10-year CSS were observed among patients with no ETE (98·6 per cent), microscopic ETE (98·3 per cent) and gross ETE invading only strap muscles (98·9 per cent) ($P = 0·375$). Microscopic ETE (hazard ratio (HR) 1·14, 95 per cent 0·84 to 2·33) and gross ETE invading only strap muscles (HR 1·49, 0·61 to 3·63) were not associated with an increased risk of death compared with no ETE. Ten-year CSS rates were lower in patients with gross ETE invading perithyroidal structures other than strap muscles (90·0 per cent) than in the other three groups (unadjusted HR *versus* no ETE 6·63, 3·65 to 12·01). Patients with gross ETE invading perithyroidal structures other than strap muscles had

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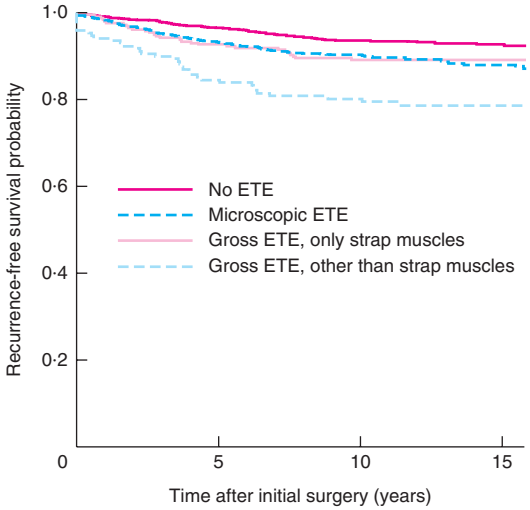
Table 4 Structural recurrence and recurrence-free survival according to degree of extrathyroidal extension

	No. of patients	No. of recurrences*	Recurrence-free survival (%)		Cox regression analysis	
			10 years	15 years	Hazard ratio†	*P*
No ETE	1362	94 (6·9)	93·7	92·8	1·00 (reference)	
Microscopic ETE	1377	153 (11·1)	90·3	88·1	1·67 (1·28, 2·14)	< 0·001
Gross ETE invading only strap muscles	261	28 (10·7)	89·2	89·2	1·67 (1·10, 2·55)	0·017
Gross ETE invading perithyroidal structures other than strap muscles	174	36 (20·7)	80·3	78·7	3·49 (2·38, 5·13)	< 0·001
Overall	3174	311 (9·8)				

Values in parentheses are *percentages and †95 per cent confidence intervals. ETE, extrathyroidal extension.



No. at risk				
No ETE	1362	1294	1156	310
Microscopic ETE	1377	1267	1130	254
Gross ETE, strap muscles only	261	240	210	16
Gross ETE, other than strap muscles	174	139	120	24

Fig. 2 Kaplan–Meier survival curves for recurrence-free survival according to extent of extrathyroidal extension (ETE). $P < 0·001$, no ETE *versus* microscopic ETE; $P = 0·016$, no ETE *versus* gross ETE invading only strap muscles; $P < 0·001$, no ETE *versus* gross ETE invading perithyroidal structures other than strap muscles; $P = 0·962$, microscopic ETE *versus* gross ETE invading only strap muscles; $P < 0·001$, microscopic ETE *versus* gross ETE invading perithyroidal structures other than strap muscles; $P = 0·003$, gross ETE invading only strap muscles *versus* gross ETE invading perithyroidal structures other than strap muscles (log rank test)

significantly worse CSS compared with the other three groups, which did not differ significantly from one another ($P = 0·201$, no ETE *versus* microscopic ETE; $P = 0·317$, no ETE *versus* gross ETE invading only strap muscles; $P = 0·778$, microscopic ETE *versus* gross ETE invading only strap muscles) (*Fig. 1*).

When patients were divided based on a tumour size cut-off of 4·0 cm, similar results were obtained for the subgroups (*Table S2*, supporting information). Outcomes were also analysed by dividing patients based on the year of initial surgery in 5-year intervals. The extent of ETE was associated with reduced CSS in the first 5-year period, but not in the later interval (*Table S3*, supporting information).

Recurrence-free survival according to extent of extrathyroidal extension

Some 311 of the 3174 patients (9·8 per cent) developed recurrence, a median of 3·9 (i.q.r. 1·7–7·0) years after initial operation (*Table 4*). Ten-year RFS for patients with gross ETE invading only strap muscles was significantly shorter than that among patients with no ETE (89·2 *versus* 93·7 per cent respectively; $P = 0·016$), and similar to

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Table 5 Results of multivariable Cox proportional hazards analyses to identify factors associated with thyroid cancer specific survival and recurrence-free survival in patients with differentiated thyroid carcinoma

	Cancer-specific survival		Recurrence-free survival	
	Hazard ratio	*P*	Hazard ratio	*P*
Extent of ETE				
No ETE	1·00 (reference)	0·232	1·00 (reference)	0·112
Microscopic ETE	1·32 (0·77, 2·26)	0·310	1·27 (0·98, 1·66)	0·076
Gross ETE invading only strap muscles	0·89 (0·35, 2·26)	0·804	1·09 (0·71, 1·69)	0·685
Gross ETE invading perithyroidal structures other than strap muscles	1·90 (0·96, 3·80)	0·067	1·60 (1·05, 2·43)	0·027
Age at diagnosis (years, continuous)	1·11 (1·08, 1·13)	< 0·001	1·00 (1·00, 1·01)	0·310
Sex (F *versus* M)	0·49 (0·30, 0·81)	0·005	0·83 (0·62, 1·11)	0·215
Histology (non-PTC *versus* PTC)	2·93 (1·30, 6·62)	0·010	1·46 (0·82, 2·61)	0·201
Primary tumour size (cm)				
≤ 2·0	1·00 (reference)	< 0·001	1·00 (reference)	< 0·001
2·1–4·0	2·44 (1·45, 4·11)	0·001	1·97 (1·52, 2·55)	< 0·001
> 4·0	7·40 (4·02, 13·63)	< 0·001	2·91 (1·92. 4·42)	< 0·001
Cervical lymph node metastasis (yes *versus* no)	1·00 (0·62, 1·61)	0·993	2·69 (2·09, 3·46)	< 0·001
Distant metastasis (yes *versus* no)	2·78 (1·33, 5·85)	0·007	8·76 (5·86, 13·10)	< 0·001
Extent of surgery (total thyroidectomy *versus* lobectomy)	2·66 (0·78, 9·15)	0·120	0·91 (0·55, 1·50)	0·718

Values in parentheses are 95 per cent confidence intervals. ETE, extrathyroidal extension; PTC, papillary thyroid carcinoma.

the 10-year RFS rate in patients with microscopic ETE (90·3 per cent). Patients with microscopic ETE (HR 1·67, 95 per cent c.i. 1·28 to 2·14) and gross ETE invading only strap muscles (HR 1·67, 1·10 to 2·55) had a significantly higher risk of recurrence than patients with no ETE. The 10-year RFS rate was lowest for patients with gross ETE invading perithyroidal structures other than strap muscles (unadjusted HR *versus* no ETE 3·49, 2·38 to 5·13). Patients with gross ETE invading only strap muscles had significantly shorter RFS than those with no ETE ($P = 0·016$), and longer RFS than those with gross ETE invading perithyroidal structures other than strap muscles ($P = 0·003$) (*Fig. 2*). RFS in patients with gross ETE invading only strap muscles was not significantly different from that for patients with microscopic ETE ($P = 0·962$).

Recurrence outcomes were also analysed by dividing patients based on a tumour size cut-off of 4 cm, and similar patterns were seen for patients with a primary tumour size of 4·0 cm or less (*Table S4*, supporting information). For patients with primary tumours larger than 4·0 cm, microscopic ETE and gross ETE invading only strap muscles were not significantly associated with recurrence. In an analysis of recurrence with patients divided based on the year of initial surgery at 5-year intervals, extent of ETE was associated with reduced RFS, except in the group with gross ETE invading only strap muscles in the first 5-year interval (*Table S5*, supporting information).

Risk factors associated with survival and recurrence

In multivariable analysis, extent of ETE was not associated with reduced CSS (*Table 5*). Factors independently related to CSS were age at diagnosis, sex, histology, tumour size and distant metastasis. No ETE, microscopic ETE and gross ETE invading only strap muscles were not associated with RFS in multivariable analysis. Only gross ETE invading perithyroidal structures other than strap muscles was an independent predictor of recurrence (HR 1·60, 95 per cent c.i. 1·05 to 2·43; $P = 0·027$). Other factors independently related to RFS were tumour size, lymph node metastasis and distant metastasis.

Discussion

Although DTC is generally considered an indolent and slowly progressing tumour, some clinicopathological features indicate poor prognosis. This study evaluated the prognostic significance according to extent of ETE, especially gross ETE invading only strap muscles, in patients with DTC. The major finding was that, although gross ETE invading only strap muscles might provide prognostic information about persistent/recurrence, it had no impact on long-term disease-specific survival.

The negative impact of gross ETE invading major structures (T4 disease in the AJCC TNM staging system) on recurrence and mortality is accepted widely[3,11,12] and was reproduced in this study. Although the difference in mortality was not statistically significant in multivariable analysis, there was a tendency towards decreased CSS in patients with gross ETE invading perithyroidal structures other than strap muscles, with an increased HR. It is speculated that widely known prognostic variables such as age, larger tumour size and distant metastases are stronger risk

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factors, and so the extent of ETE was not an independent predictor of survival.

According to the present data, gross ETE invading only strap muscles was not related to disease-specific mortality, even in univariable analysis. This finding was similar to results from previous studies[8,13,14] on the prognosis of minimal ETE. In the sixth and seventh editions of the AJCC TNM staging system[4,15], T3 disease includes any tumour with minimal ETE (such as extension to sternothyroid muscles or perithyroidal soft tissues). Several studies have attempted to clarify the impact of minimal ETE on the prognosis of DTC. There is no strong evidence that minimal ETE predicts CSS, and most studies have suggested that it is not an independent predictor of mortality. Although some studies[7,16,17] reported minimal ETE as a risk factor for recurrence and decreased RFS, others[5,6] found that it was unrelated to poor outcome.

For this reason, in the eighth edition of the AJCC TNM system[10], gross ETE invading only strap muscles replaces minimal ETE as a determinant of T3b disease. As validated in several reports[18,19], the revised eighth TNM system is superior to the seventh edition because of the remarkable changes with regard to age cut-off point, T descriptions and stage groups. However, the prognostic significance of subdividing T3 into T3a and T3b has not been validated. Evidence is still lacking for the prognostic significance of gross ETE invading only strap muscles. Here, it was demonstrated that, like minimal ETE, gross ETE invading only strap muscles did not affect disease-specific mortality and was not an independent predictor of recurrence.

ETE invading only strap muscles could be associated with recurrence. Although patients with gross ETE invading only strap muscles and microscopic ETE may have worse recurrence outcomes than patients with no ETE, current evidence indicates that ETE, particularly that confined to the anterior neck compartments[18,20], is not useful as an independent predictor of survival or recurrence. Gross ETE invading only strap muscles had a prognostic value similar to that of microscopic ETE in the present study. Microscopic ETE and gross ETE invading only strap muscles were associated with CSS similar to that of no ETE, and both had worse RFS. Similarly, in a previous study[5] that divided patients with minimal ETE into strap muscle invasion-positive and -negative subgroups, no significant differences were seen in recurrence or survival according to degree of strap muscle invasion. This finding might be related to easier resection of gross ETE invading only strap muscles, which is in line with a previous study[13] demonstrating that ETE in the posterior direction has worse RFS than ETE in the anterior direction. When negative margins, another well known prognostic predictor, are obtained, the degree of direct invasion might, however, no longer significantly affect long-term prognosis.

In the past 20 years, radioactive iodine (RAI) adjuvant therapy has commonly been used with patients with minimal ETE. Based on lack of a clear benefit of RAI for patients with minimal ETE, the 2015 American Thyroid Association guidelines[21] suggest that smaller tumours with microscopic ETE may not require RAI. Although the impact of RAI therapy on recurrence and survival in patients with gross ETE invading major structures is well described in observational studies, there are insufficient data to mandate use of RAI in patients with gross ETE invading only strap muscles. Appropriate prediction of prognosis for patients with gross ETE invading only strap muscles is also important for balancing the benefits and risks in decision-making regarding use of postoperative radioiodine.

Strengths of the present study are the large cohort and long follow-up. A limitation is the retrospective study design and relatively small numbers of patients with gross ETE invading only strap muscles and gross ETE invading perithyroidal structures other than strap muscles. In addition, heterogeneity in the prognostic impact of gross ETE over time was found. Temporal changes in the clinicopathological characteristics of tumours and surgical practice, and improved diagnostic tools and management strategies, might be responsible for this result, and should be studied further[22,23].

This study has demonstrated that gross ETE invading only strap muscles does not affect disease-specific mortality and is not an independent predictor of structural recurrence. The actual impact of gross ETE invading only strap muscles will be important in revising the staging system in the future, as well as for better prognostication in individual patients.

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S.Y.P. and H.W.J. contributed equally to this study.
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